THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In 1202_3, the value (in decimal) contributed by the digit 2 in position 2 from the left is: 1

(a) 18 **(b)** 6 **(c)** 11 **(d)** 54

2. The decimal value of 1431_5 is: 1

(a) 243 **(b)** 239 **(c)** 247 **(d)** 241

3. The binary form of 62 is: 1

(a) 011111 **(b)** 111111 **(c)** 111110 **(d)** 111101

4. CLXXXIX + CCCXI equals: 1

(a) D **(b)** DI **(c)** CDXC **(d)** DX

5. The largest 3-digit number in base 6, written in decimal, is: 1

(a) 215 **(b)** 216 **(c)** 35 **(d)** 17

6. The number of digits in the binary form of 128 is: 1

(a) 7 **(b)** 9 **(c)** 10 **(d)** 8

7. $1110_2 + 11_2$ equals: 1

(a) 10001_2 **(b)** 10000_2 **(c)** 10011_2 **(d)** 10010_2

8. In 45_6, the value (in decimal) contributed by the digit 4 in position 1 from the left is: 1

(a) 144 **(b)** 24 **(c)** 10 **(d)** 25

9. The decimal value of 374_8 is: 1

(a) 254 **(b)** 258 **(c)** 252 **(d)** 250

10. The binary form of 29 is: 1

(a) 10111 **(b)** 11100 **(c)** 11101 **(d)** 11110

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Subtract $101111_2 - 10010_2$ and give the answer in binary. 2

12. Convert 5163_8 to decimal. 2

13. Write CCCXLVIII + CCLXVII as a Roman numeral. 2

14. Subtract $111010_2 - 10111_2$ and give the answer in binary. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 110011_2 to base 4 by first converting to decimal. 3

16. Multiply $1011_2 \times 11_2$ and give the answer in binary. 3

17. Write 2450 and 899 in Roman numerals and find their difference in Roman numerals. 3

OR

Convert 243_6 to base 8 by first converting to decimal. 3

18. Multiply $110_2 \times 111_2$ and give the answer in binary. 3

OR

Write 1999 and 749 in Roman numerals and find their difference in Roman numerals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the sum of 11110_2, 1011_2 and 111_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 3-digit numbers in base 6 and find their decimal values. How many 3-digit numbers are there in base 6? 5

20. Find the sum of 10011_2, 1111_2 and 1000_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 3-digit numbers in base 7 and find their decimal values. How many 3-digit numbers are there in base 7? 5

END OF QUESTION PAPER — Total: 40 Marks

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